

Data and Phenomena - A restatement and defense

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Author's Motivating Questions

Provides an expansion and generalisation of the ideas introduced in Bogen and Woodward (1988), while responding to some criticisms stated against the original paper, specifically:

1. There is a common view that scientific theories should (systematically) explain the data that is used in theory - theory-mediated data. The 1988 paper argues that this is the wrong direction of explanation for understanding the evidence empirical observations provide and the inferences conducted in science to reason from observation to phenomena.
2. How does theory remain relevant in our understanding of the inference from data to phenomena? And what additional background assumptions are needed to make sense of this inferential reasoning?
3. Bogen and Woodward vs scepticism (e.g., McAllister 1997): Defends their original distinction between data and phenomena from worries that data-to-phenomena inference requires theory to explain the data or that the role of goals and interests of the theorist making the inferences makes the phenomena relative.

Woodward's thesis: the role of data is to serve as evidence for the existence of phenomena, and scientists do not need theories to systematically explain the data in order for this inferential reasoning to hold between data and phenomena.

Definitions:

Explanatory theory: theories that provide detailed and systematic explanations of what they call '*phenomena*'. These types of explanations are commonly in the form of derivations from basic laws and principles

Singular causal explanation: a specification (within a theory) that some causal factor plays a certain role in the production of an outcome. These types of explanation do not specify in a 'detailed and quantitative way' how the outcome depends on the causal factor(s).

Data: records that are produced by measurement and experiment, which serve as empirical evidence for the existence of certain phenomena.

Phenomena: features of the world that, in principle, could recur under different contexts or conditions.

The argument:

1. There is a data/phenomenon distinction in scientific theories

Scientific investigations do not tell us about the phenomena alone: the design of experiments is such that the data produced will reflect features of the phenomena under investigation and other causal factors, e.g., non-generalised factors that are specific to human behaviour, or to the measurement apparatus and the design of the experiment and the local environment.

Example: consider the measurement of a fixed quantity M . If the measurement procedure is followed exactly, the data d_i will be observed at different measurement occasions. This can be characterised as a function f of M , with some additional causal factors u_i that specifies the differences in the value observed on different occasions.

- It follows from this that: $d_i = f(M, u_i)$
- The measurement of value M is attainable from the scientific investigation, but the additional causal factors u_i will not be quantifiable from the measurement - they will have to be accounted for by the additional assumptions the theorists believe to be satisfied in this investigation.

Explanatory theories provide explanations of phenomena - they reason from theory-mediated data to phenomena, and in so doing, they need not, nor is it necessary that the theory provide an explanation of the data that is used to infer the phenomena from the scientific explanation.

Phenomena need not be observable - many discussions of the role of observation in science do not consider how phenomena are detected, as this has little to do with how human perception works, but instead on inferential reasoning from data to phenomena.

Importantly, these conclusions are not intended to replace the "top-down" view of scientific theories. They do not intend to argue that science should be considered in a singular way: top-down or bottom-up. Instead, their aim was to

1. Provide a framework that makes sense of various sorts of data-based reasoning, one that grants that such reasoning from data to phenomena can occur and be relatively independent of certain kinds of theories.
2. Show that this framework can sometimes play a more leading role than has been previously thought.

Overall, however, there is no singular way to do scientific research; the approach, theory-driven or data-driven, should depend on (a) what the problem is, (b) what sort of information is available and (c) what questions the scientist wants to answer.

2. Data-to-phenomena reasoning requires background assumptions

Data-to-phenomena reasoning is a type of inductive reasoning; thus, the derived conclusion about a given phenomenon goes beyond the data on which it is based - *additional substantive empirical assumptions* are always required:

Evidence + substantive empirical assumptions → conclusion (phenomena)

These empirical assumptions play specific roles:

- They help *determine which features of the data count as relevant* signals of the phenomena and which are background causal factors that need to be accepted as 'unexplained' variance in the data.
- In having a source independent of the data generated in the particular experiment or context observed, they are *original assumptions that are either true or false* of the experimental context at hand.
 - E.g., are the different causal factors contained in u_i independent from each other and are additive, such that the central limit theory can be employed to assume a normal distribution of error present in the calculation of M ?
- *There is no single universal assumption* that can be applied across all cases: different (true) assumptions can license different claims from the same data obtained about a given phenomenon.
 - McAllister's scepticism: McAllister's notion of a "noise level" that is dependent on the investigator's cognitive interests and attitude towards risk. This leads to the inference of phenomena that are stipulative or subjective
Woodward's response:
 1. This need for additional assumptions and stipulations to infer from data to phenomena is a feature of all inductive inference
 2. What the investigator's cognitive interests and attitudes affect is (1) the assumptions that they utilise in their reasoning from data to phenomena. Or (2) what investigative questions they choose to undertake. And if that assumption is true, then it will license a true and consistent claim about the phenomena from the data, not an inconsistent claim. Either way, there is no subjective or stipulative inferential reasoning.

3. Data-to-phenomena reasoning does not require a theory that systematically explains the data

Explanatory theory can matter in data-to-phenomena reasoning, without explaining the data that allows for the reasoning in a systematic way. The roles the theory can play are:

1. Provide *motivation* for why a phenomenon is worth measuring; to test the theory, or because the phenomenon is an important feature of the theory that explains a fundamental feature of nature.

2. Provide the researcher with the *vocabulary for describing the results* of measuring the phenomenon; this is only possible if the theory is considered seriously
 - Importantly, when a theory plays this role in the data-to-phenomena reasoning, it does not entail circular reasoning: the data does not enforce the theory's truth; the theory can still be falsified.

What the theory allows for is the vocabulary to describe, in a systematic way, the derivations that allow for the phenomena to occur (if the results predicted by the theory are obtained).

1. *Theory can reinforce our confidence in the predicted phenomena and vice versa.* By predicting the reality of the phenomena, the theory can increase our confidence that the experiment is reliable and that potential errors have been controlled for. This increases our confidence in the reality of the phenomena, particularly in cases of limited evidence. Similarly, in cases where the theory is not yet accepted, the successful detection of the phenomena gives us good reason to accept the theory that predicted it.

- Importantly, this positive feedback is not circular because the phenomenon is not inferred from the theory alone. It must already be supported by data and background assumptions, which remain the decisive and constraining basis for its acceptance. The theory's successful prediction then provides additional, independent support, strengthening confidence in a claim that is already data-supported, especially under conditions of limited evidence.

To do all of this, the theory need not provide a detailed explanation, i.e., a derivation from first principles, of how each measurement outcome is obtained.

Key Takeaways

This paper is not a radical argument for how evidence should be understood universally across the whole of 'science'. Instead, it is a defence, a practical account of how phenomena can be reliably extracted from theory-mediated data, grounded in how science is actually observed to be done.

Data-to-phenomena reasoning is an inferential move that does not require a theory to provide a systematic explanation of the data. The direction of reasoning here goes from data to the phenomena the researcher is interested in. This is useful in our scientific practice, given the data-rich, theory-poor position some scientific fields occupy.

The inferential reasoning in data-to-phenomena is not "from data alone". It also relies on substantial empirical assumptions that effectively connect the data to a phenomenon.